Introduction

What PGMs are and how this course is organised

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This class is a conceptual introduction to probabilistic graphical models (PGMs), it also covers course organisation.

ILOs After this class the student

- understands what PGMs are about at a conceptual level;
- recognises what PGMs are meant for;
- knows what this course will cover;
- knows how this course is organised.

Textbook for this course: Koller and Friedman [3].

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Probabilistic Graphical Models

Probabilistic Graphical Models (PGMs)

Probabilistic graphical models combine

- statistics and probability theory
- and computer science

to solve hard problems involving variables we are **uncertain** about.[2]

[2] *Uncertainty* concerns a state of imprecise or incomplete knowledge, where, given the information available, that which is possible cannot be reduced to a singleton [1].

Early applications

Medical diagnosis: uncertainty about predisposing factors, symptoms, test results, diseases, treatment outcomes, etc.

Image segmentation: uncertainty about the class of pixel (or super-pixels) e.g., 'sky', 'water', 'grass', 'cow', 'flight', etc.

Translation: uncertainty about bilingual dictionary entries and transfer rules.

What do these problems have in common?

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What do these problems have in common?

- large number of variables,
- significant uncertainty about the right answer.

Probabilistic Graphical Models

The **model** is a declarative representation of our understanding of the world.

It is expressed in terms of the *variables we are reasoning about* and how we believe or assume they interact with one another.

PGMs are designed to help us **represent our uncertainty** and **support decision making**.

Uncertainty comes in many forms and for many reasons:

- Partial knowledge of the state of the world
- Noisy observations
- Modelling limitations
- Inherent ambiguity and/or stochasticity

Probability theory is the most widely used framework for uncertainty representation.

For other mathematical frameworks for uncertainty repreentation see Halpern [2].

Graphs allow us to represent complex systems involving many interacting variables.

We will use graphs to code domain knowledge and represent complex probability distributions.

Graphs are **efficient data structures** that support **efficient algorithms** and are amenable to **systematic analysis**: we can ascertain properties of distributions by studying the graph structure.

Summary

What is a probabilistic graphical model?
A representation of the world that takes into account the things we care and are uncertain about and our assumptions about their relationships.

What can we do with it?
Answer queries about the world and make decisions under uncertainty.

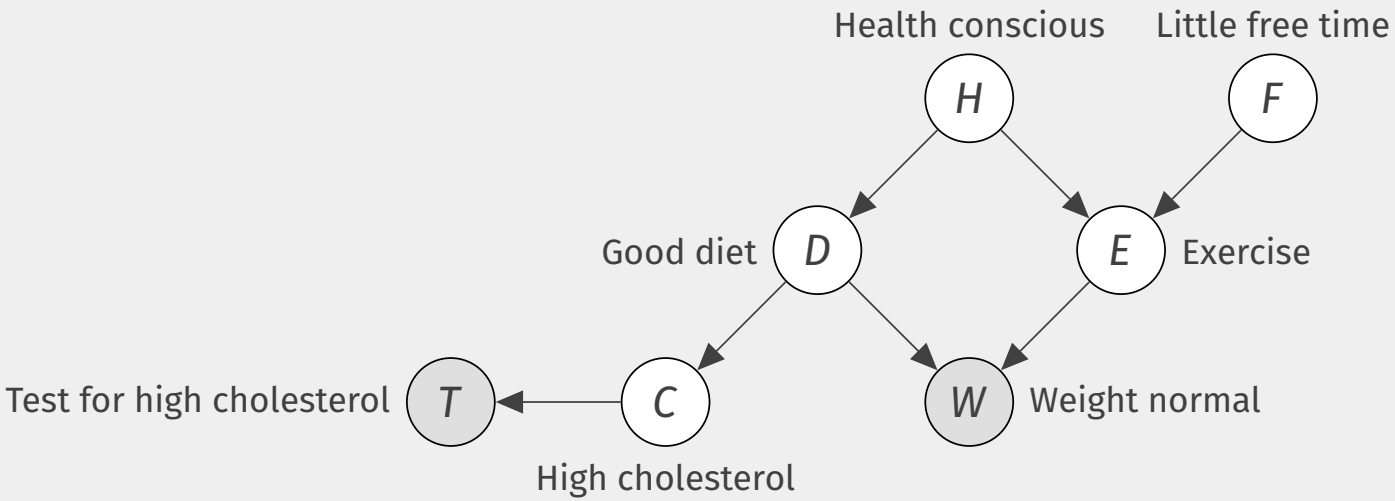
How do we construct them?
A combination of domain knowledge and learning from historical data.

Widespread Applications

- Medical diagnosis
- Fault diagnosis
- Natural language processing
- Traffic analysis
- Social network models
- Message decoding
- Computer vision
- Speech recognition
- Robot localisation and mapping
- Forecasting (weather, finances, supply chains, etc.)

and the list goes on...

Examples



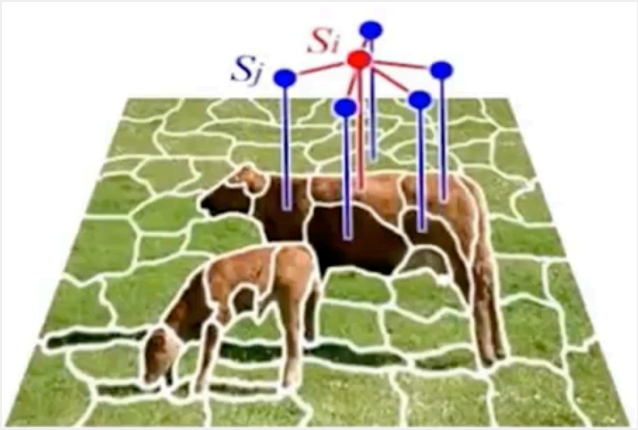
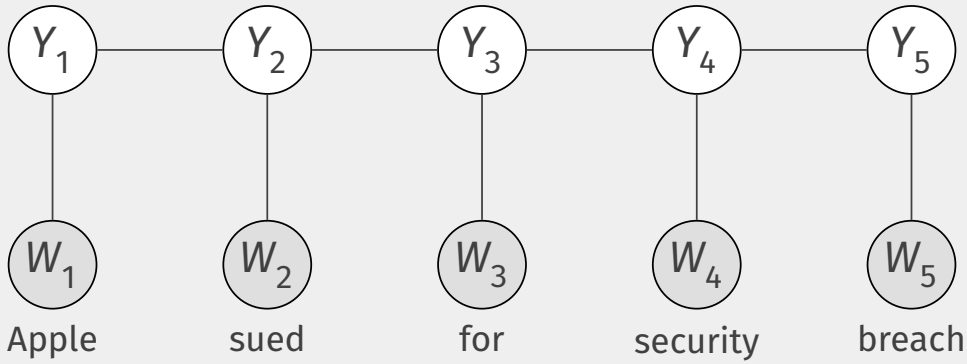


Image by Daphne Koller



What are you going to learn?

Representation

- Directed and undirected graphical models
- Parameterisation

Reasoning

- Exact and approximate inference
- Decision making

Learning

- Parameters
- Observed and unobserved data

Course Organisation

Platforms

The course is managed entirely on Canvas: official source of information, announcements, deadlines, material and class recordings: `https://canvas.uva.nl/courses/59709`

We use a few additional tools, for specific purposes:

- ANS: exam grading and feedback `https://ans.uva.nl`
- Ed: asynchronous Q&A (see Ed House Rules on Canvas) `https://canvas.uva.nl/courses/59709/pages/ed-house-rules`

Learning Activities

HCs: we develop the main theory planned for the week and solve some exercises; sometimes there's reading that's necessary before or after the class.

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LCs: we translate the main ideas in PGMs to Python code; because PGMs deal with graph structure, probabilistic inference and statistical learning, programming exercises will often help you consolidate your knowledge of the theory.

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WCs: we work on exam-like exercises, discuss feedback, and sometimes cover some side topics that help you get a good hold of the material.

Assessment

Exams: midterm (40%) and endterm (40%).

Programming: weekly assignments (due on Monday), the best 4 (out of 6) contribute to 10% of your final grade.[4]

Exam-like questions: weekly assignments *at the weekly Werkcollege*, the best 4 (out of 6) contribute to 10% of your final grade.[5]

In-class bonus: occasionally, we may offer in-class assignments in an Hoorcollege; collectively, these assignments contribute up to *1 bonus point*; the bonus goes directly to your final grade (see next).

[4] Submission in pairs through Canvas. Register your groups asap.
[5] Individual submission (on-site).

Grade

Your **final grade** is obtained as follows:[6]

$$= \text{SIS}(0.8 \times \text{exam} + 0.1 \times \text{programming} + 0.1 \times \text{werkcollege} + \text{bonus})$$

No minimum requirement on any one of the partial grades, but the *final grade* must be a 6.0 for you to pass this course.

See `https://canvas.uva.nl/courses/59709/pages/assessment-and-grade`

[6] SIS rounds and clips the grade according to fixed rules, see Canvas.

What Next?

Rest of HC1a: BNs (semantics)

LC1: BNs in code

HC1b: BNs (reasoning and influence)

WC1: BN exercises (semantics, reasoning and influence)

References I

[1] Didier Dubois and Henri Prade. Formal representations of uncertainty. In *Concepts & Methods of Decision-Making*, chapter 3. John Wiley & Sons Inc., 2009.

[2] Joseph Y Halpern. *Reasoning about uncertainty*. MIT press, 2017.

[3] Daphne Koller and Nir Friedman. *Probabilistic graphical models: principles and techniques*. MIT press, 2009.